

/Y9 CA1 Revision

Topics to study: Genes and Inheritance; Periodic Tble

Keywords	Definitions
Neutron	Neutral particle in the nucleus of an atom
Homozygous	Having 2 identical alleles of a particular gene
DNA	The hereditary molecule inside cells that carries the genetic instructions used in the development, functioning, and reproduction of all living
Heterozygous	When two people have 2 different alleles for the same gene. The dominant allele will show its traits
Recessive allele	An allele that will only show its traits when two of them are present. Will be shown in two lowercase letters.
Dominant gene	A version of a gene that masks or overrides the effect of a different version (the recessive allele) on the other chromosome
electron	A negatively charged particle in an atom(-1)
Proton	A proton is a stable subatomic particle with a positive electrical charge that lives in the nucleus (center) of every atom
compound	a substance made of two or more different elements joined together by chemical bonds in a fixed ratio
gene	The basic physical and functional unit of heredity made up of DNA.
chromosome	Thread-like structure inside the nucleus made of tightly coiled DNA and proteins.
inheritance	Inheritance means passing traits from parents to their offspring through genes.

Phenotype	The physical appearance of someone based on their genotype
Genotype	The combination of alleles a person has
	Different forms of a gene (eg: gene for a hair colour can have 3 alleles: brown, red, blond)
Electron shell	It's the outermost layer. It can hold a certain number of electrons , The number of electrons is determined by the number on the periodic table.
genes	
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Variation	The difference between individual organisms belonging to the same species.
Punnet square	A chart that shows all the possible combinations of alleles that can result from a genetic cross
halogen - group 7	The word halogen means "salt-former" in Greek because they react with metals to produce wide-ranging salts. These are elements that react with metals and have a valence electron of 7 . Their ions will have a negative charge.
alkali metals- group 1	metals that react violently with water, since their outer shell needs to remove one electron. They get more reactive down the periodic table, since there is lesser and lesser energy when you go further from the nucleus. Their ion has a positive charge. They have a valence of 1.
noble gases	a type of element that doesn't react with anything since their outer shell is full, making them really stable.
Pedigree	A chart used to trace family genetics history and inheritance patterns.
Genetics	the study of heredity
Purebreed	Having two of the same genes for a trait; another word for homozygous

Mutation	change in a DNA sequence that affects genetic information
atom	An atom is the basic building block of all matter in the universe.
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